

OAK RIDGE, TN, USA

WMO: 724270

Lat: 36.023N

Lon: 84.234W

Elev: 274

StdP: 98.07

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53868

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.4	-6.1	-16.4	0.9	-5.0	-14.0	1.2	-3.2	6.7	11.3	5.6	8.3	1.2	60	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	33.8	23.2	32.7	23.1	31.7	22.8	25.2	30.5	24.6	29.7	24.1	29.0	2.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.2	27.3	23.1	18.4	26.6	22.6	17.9	26.2	78.1	30.4	75.6	29.8	73.4	29.2	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.5	4.7	4.1	DB	-11.8	35.4	2.6	1.8	-13.7	36.7	-15.2	37.8	-16.7	38.8	-18.6	40.1	(4)
			WB	-12.6	26.0	2.4	0.7	-14.4	26.4	-15.8	26.8	-17.2	27.2	-18.9	27.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	3.6	6.0	10.5	15.8	20.3	24.3	25.8	25.6	22.3	15.9	9.8	5.6
	DBStd	8.89	5.71	5.15	5.09	4.38	3.72	2.38	2.06	2.09	3.30	4.49	4.63	4.94
	HDD10.0	624	211	131	57	7	0	0	0	0	0	7	59	153
	HDD18.3	1933	457	346	246	98	25	1	0	0	8	101	258	393
	CDD10.0	2628	13	19	72	180	318	428	489	482	368	189	52	18
	CDD18.3	896	0	1	3	22	85	178	231	224	126	25	2	0
	CDH23.3	7766	0	3	37	242	732	1547	2029	1974	994	198	10	0
	CDH26.7	2948	0	0	3	42	217	607	836	829	370	43	1	0
Wind	WSAvg	1.3	1.5	1.6	1.8	1.7	1.3	1.1	1.1	0.9	0.9	0.9	1.1	1.3
Precipitation	PrecAvg	1394	121	115	145	112	125	114	137	91	99	76	112	136
	PrecMax	1980	244	372	311	356	302	283	490	231	225	176	274	321
	PrecMin	986	24	21	57	22	20	14	39	32	35	0	45	17
	PrecStd	256	53	62	64	66	67	65	86	46	50	46	51	67
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.7	23.1	26.6	29.9	32.1	34.4	35.2	35.4	34.1	30.2	24.8	20.9
		MCWB	15.4	16.5	16.5	18.7	21.7	22.8	24.2	23.4	21.4	21.6	17.4	16.1
	2%	DB	17.1	19.7	24.1	27.9	30.5	32.9	33.5	33.7	32.3	27.6	21.9	18.0
		MCWB	13.5	14.8	15.4	17.6	20.6	22.8	24.0	23.4	21.7	19.4	15.5	14.4
	5%	DB	14.5	17.0	21.7	26.3	29.2	31.6	32.3	32.5	30.7	25.5	19.6	15.5
		MCWB	11.4	11.8	14.1	16.7	20.1	22.3	23.7	23.3	21.7	18.3	14.5	12.8
	10%	DB	12.3	14.2	19.2	24.3	27.6	30.2	31.1	31.1	28.8	23.4	17.6	13.4
		MCWB	9.6	10.1	12.8	16.0	19.6	21.8	23.3	23.0	21.0	17.5	13.3	10.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.6	18.4	18.0	20.6	23.4	25.2	26.2	26.1	24.6	23.6	19.5	17.7
		MCDB	19.1	20.7	23.5	27.4	28.2	30.7	31.7	31.1	29.5	27.4	21.9	19.6
	2%	WB	14.3	15.7	16.8	19.4	22.3	24.3	25.3	25.1	23.6	21.6	17.7	15.1
		MCDB	16.6	19.3	21.9	25.4	27.8	29.8	30.8	30.3	28.3	24.5	20.2	16.9
	5%	WB	12.5	13.1	15.6	18.3	21.5	23.7	24.7	24.5	22.8	20.0	16.0	13.4
		MCDB	14.0	15.4	20.1	23.4	26.9	28.9	29.9	29.5	27.2	23.8	18.5	15.1
	10%	WB	10.1	10.9	14.1	17.3	20.7	22.9	24.1	23.9	22.1	18.6	14.1	11.2
		MCDB	11.8	13.0	18.1	21.9	25.5	27.8	29.0	28.7	26.4	22.0	16.8	13.0
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.2	11.5	12.5	11.7	11.0	10.1	10.6	11.2	11.9	11.5	9.3
		MCDBR	11.2	13.8	14.6	15.1	13.3	12.8	11.6	12.3	13.2	14.1	14.0	11.1
		MCWBR	7.9	8.6	7.5	6.5	5.0	4.3	3.8	3.8	4.3	6.2	8.0	7.5
	5% WB	MCDBR	8.8	10.8	12.0	11.9	11.2	10.4	10.0	10.2	10.5	10.6	10.1	9.2
		MCWBR	7.1	8.3	7.2	6.0	4.6	4.1	3.7	3.8	4.0	5.5	7.1	7.4
Clear-Sky Solar Irradiance	taub	0.311	0.319	0.351	0.401	0.453	0.487	0.519	0.528	0.422	0.366	0.330	0.312	
	taud	2.510	2.479	2.423	2.277	2.157	2.099	2.039	1.991	2.274	2.427	2.490	2.527	
	Ebn at Noon	890	923	917	883	837	806	777	760	831	856	858	865	
	Edh at Noon	81	94	108	132	151	160	169	174	125	98	82	75	
All-Sky Solar Radiation	RadAvg	2.24	2.81	3.97	5.13	5.69	6.10	5.75	5.42	4.62	3.66	2.65	1.94	
	RadStd	0.20	0.36	0.37	0.35	0.43	0.39	0.35	0.27	0.43	0.46	0.29	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)		+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+138	+84

Nomenclature: See separate page